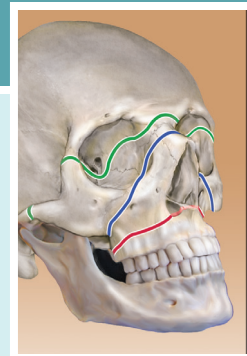


Sports and Epilepsy

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Epilepsy is a common clinical condition characterized by recurrent, unprovoked seizures. It affects approximately 1% of the general US population, with the highest incidence in infancy and childhood.^{1,2} Although previous recommendations precluded sports participation for persons with epilepsy,³⁻⁵ it is now recognized that exercise is part of a comprehensive treatment plan for epilepsy and that participation in sports and exercise should be encouraged.^{6,7}

TERMINOLOGY AND CLASSIFICATION

Seizure

An epileptic seizure is the clinical manifestation of an abnormal, hypersynchronous discharge of a group of neurons in the cerebral cortex. The clinical manifestations are determined by the seizure's anatomic focus within the brain and can include an altered state of awareness, stereotyped or repetitive movements, loss of muscle tone (atonic), tonic or clonic movements, sensory (e.g., visual) or psychic experiences, or autonomic dysfunction.

Seizures are classified as either generalized or focal. A generalized seizure is the result of rapid activation of bilaterally distributed networks, which include both cortical and subcortical structures. Generalized seizures can be either convulsive (e.g., tonic, clonic, tonic-clonic, myoclonic, or atonic) or nonconvulsive (e.g., absence). Focal seizures, previously classified as partial seizures,⁸ involve a limited network of neurons localized to a region of the brain. Although seizures traditionally have been subdivided into simple focal or complex focal categories, based on altered (simple) or impaired (complex) consciousness, it has been proposed⁹ that the traditional terminology be replaced with a description of the elemental features of the seizures and the sequence of occurrence.^{10,11}

Simple focal seizures can include motor, sensory, autonomic, and psychic phenomena. These seizures, often referred to as an "aura," can transform into a complex focal seizure as the seizure focus expands to involve other regions of the cortex that mediate awareness. A seizure that starts in the temporal lobe may initially begin as a simple focal seizure (aura) of a rising gastric sensation followed by staring, diminished responsiveness, and stereotypic automatisms including repetitive swallowing or lip smacking (a complex focal seizure). Focal seizures can evolve into a generalized tonic-clonic seizure, an event referred to as "secondary generalization."

Epilepsy

Epilepsy, which is defined as a condition of unprovoked, recurrent seizures, can be either inherited or acquired. Inherited or genetic forms of epilepsy (previously called "idiopathic" epilepsy) are often channelopathies, that is, disorders that alter the excitability of the neurons, resulting in epilepsy. Examples include sodium channelopathies and potassium channelopathies. Other types are "symptomatic," implying an underlying central nervous system lesion/abnormality, such as encephalomalacia due to trauma, stroke, infection, or cortical dysplasia. Symptomatic epilepsies are generally difficult to treat and usually do not go into remission.

An epilepsy syndrome is defined by the seizure type(s), age of seizure onset, developmental and family history, electroencephalogram (EEG) and neuroimaging results, neurologic examination, and medication response. Examples of epilepsy syndromes include childhood absence epilepsy, juvenile myoclonic epilepsy, and benign rolandic epilepsy.

The epilepsy syndromes were previously subdivided into two major categories: generalized epilepsy syndromes and localization-related (or focal) epilepsy syndromes.¹² However, an expanding knowledge of epilepsy recommends excluding the prior classification system in favor of a system that separates the manifestations of epilepsy from the pathogenesis. In the newly proposed system,⁹ syndromes are subdivided into pure electroclinical syndromes, constellations, structural/metabolic epilepsies, and epilepsies of unknown causes.

DIAGNOSIS AND EVALUATION

The diagnosis of epilepsy is based on clinical history. A single seizure can be provoked by electrolyte imbalances, dehydration, infection, or trauma. Other conditions should also be considered in persons presenting with a paroxysmal event, including cardiac arrhythmias, prolonged QT syndrome, syncope, complicated migraine, paroxysmal movement disorders, and psychogenic events. Although an outpatient EEG is an indispensable tool in clinical diagnosis, clinical history is the most critical determinant. Of all persons with epilepsy, 17% have normal interictal EEG findings (when they are not having an overt seizure). In addition, abnormal EEG findings do not guarantee that a seizure has occurred.¹³⁻¹⁵ Lastly, following the diagnosis of epilepsy, a head

Abstract

Epilepsy is a common condition that affects approximately 1% of the general population, with the highest incidence in infancy and childhood. Defined as recurrent, unprovoked seizures, epilepsy can be either inherited/genetic or acquired/symptomatic, implying an underlying central nervous system abnormality. The evaluation typically includes an electroencephalogram and a magnetic resonance imaging scan of the brain. With recurrent seizures, an antiepileptic medication is usually prescribed. Approximately 70% of patients with epilepsy will be well controlled using antiepileptic medications. Past recommendations discouraged sports participation for persons with epilepsy. Studies have now demonstrated that sports participation and exercise do not increase seizure frequency and may actually be “seizure antagonists.” Furthermore, exercise typically decreases the common comorbidities of epilepsy, including anxiety, depression, low self-esteem, and peer isolation. Persons with well-controlled epilepsy may have a seizure during exercise, but this is rare and generally self-limited. Furthermore, epilepsy should not automatically exclude a person from exercise or sports participation. In fact, persons with well-controlled epilepsy should be encouraged to exercise. In making this decision, several factors need to be considered: seizure type and frequency; type of sports; seizure precipitants; and the individual’s personal need to participate.

Keywords

epilepsy and sports
epilepsy and exercise
epilepsy and contact sports

imaging study—preferably a magnetic resonance imaging scan of the head—should be obtained. Exceptions to this rule may include a child or an adult diagnosed with childhood absence epilepsy, juvenile absence epilepsy, or juvenile myoclonic epilepsy.¹⁶

TREATMENT

Therapy with an antiepileptic drug (AED) is not typical after a single seizure because recurrence risk after a first unprovoked seizure is only 30% to 40%. Risk for additional seizures increases to approximately 75% after a second unprovoked seizure.^{17–19} After this second seizure, an AED is usually prescribed. Occasionally, an AED is started after a first unprovoked seizure if the EEG demonstrates epileptiform discharges (spike or sharp waves) in the temporal region or generalized spike and slow wave discharges. The chance of recurrence in these situations may be closer to 90%.

Approximately 30 AEDs are used to treat epilepsy. The choice of AED is based on seizure type and epilepsy syndrome. The following general principles relate to the choice of AED:

1. Monotherapy is effective and avoids unpleasant drug interactions. About 60% of patients will experience control of their epilepsy with an appropriate first AED. An additional 10% will respond to a second AED. Unfortunately, chances of a subsequent medication trial succeeding when the first two AEDs have failed are less than 5%.^{20,21} Thus 30% of patients will most likely be refractory to therapy. For these patients, alternative therapies including vagus nerve stimulation, diet therapies (e.g., a ketogenic diet), and epilepsy surgery need to be considered.
2. If possible, AEDs should be titrated slowly and only to the point of seizure control.
3. Seizure control should not be achieved at the expense of adverse effects. An alternative medication should be chosen if a person experiences adverse effects.
4. Drug compliance is enhanced when medication is administered once or twice daily. When available, a sustained-release medication should be considered.
5. Therapeutic blood levels are not absolute; they are formulated based on trough levels and represent a statistical range of efficacy.

Persons caring for patients with epilepsy should be cognizant of AED adverse effects that may impair sports participation. For example, some AEDs at high doses can produce cognitive or behavioral changes, diplopia, dizziness, and general fatigue. Phenytoin, carbamazepine, valproate, and lamotrigine can induce tremors or dyskinesias.^{22,23} Topiramate and zonisamide can cause oligohydrosis.^{24,25} Thus patients should be advised to maintain hydration, carry a spray bottle of cool water, and monitor for overheating when participating in sports. Carbamazepine and oxcarbazepine can cause hyponatremia.²⁶ Additionally, valproate can cause weight gain and topiramate can induce weight loss.^{27–29}

Several AEDs can affect bone health by decreasing bone mineral density in adults and children.³⁰ One study demonstrated that as a group, people with epilepsy had lower overall bone density

than did the general population.³¹ Furthermore, the bone density of persons for whom an enzyme-inducing AED was prescribed was lower than that of persons for whom a nonenzyme-inducing agent was prescribed. Therefore monitoring bone health is critical in patients with epilepsy, especially those using phenytoin, phenobarbital, or carbamazepine. Both the patient and health care provider should be aware of the risk of bone fractures as a result of sports participation.

EXERCISE AND SEIZURES

In 1960, Dr. William G. Lennox, an early leader of pediatric epileptology, wrote: “Epilepsy prefers to attack when the person is off guard, sleeping, resting, idling. This is easily demonstrable in persons who experience very frequent petits [absence seizures], which may be almost absent during skating, swimming, or running, and abundant while sewing, eating, or just sitting. Parents picture their child as stopping or falling in a seizure and being run over while attempting to cross a busy street. I have never known this to happen, and I can remember only a few instances of a person having an attack while running or swimming.”³² Based on these observations, Lennox labeled activity as an *antagonist of seizures*.

Most human and translational animal studies completed in the interim support Dr. Lennox’s empirical conclusion that exercise may, in fact, be protective against seizures and that it is rare for seizures to occur during activity. However, it is reasonable to expect that physical activity may induce seizures because strenuous exercise can result in hyperventilation, and in the EEG laboratory, hyperventilation is used to provoke seizures. However, the net effects of hyperventilation at rest and during exercise are different. At rest, hyperventilation triggers a decrease in carbon dioxide (respiratory alkalosis). The ensuing drop in cerebral blood flow and hypoxia results in network excitability and seizures. During exercise, however, hyperventilation is a compensatory mechanism that avoids hypercapnia and meets increased oxygen demands, thus the respiratory alkalosis required to induce a seizure does not occur.³³

In addition, the most common provoking factors for seizures include stress, mental strain, and physical fatigue,³⁴ all of which can occur with intense physical exercise and competition. Stress may activate seizures through sympathetic stimulation,³⁵ and fatigue may cause seizures because of a chronically drowsy state.³⁶ Metabolic disturbances during prolonged, strenuous exercise, such as hyponatremia, dehydration, and overhydration, can theoretically aggravate an underlying propensity for seizures. However, the literature includes only a few cases of clear exertion-induced seizures,^{37–40} and larger studies have not documented exercise-induced seizures, even during contact sports.⁴¹

Several studies and observations bear mention. In a single-center study of more than 15,000 patients with epilepsy, not a single case of exercise-induced seizures was reported.⁴² In a second observational study of 400 patients, only two patients reported seizures during exercise.⁴³ Nakken⁴⁴ has reported that 10% of 204 patients with epilepsy had minor injuries during exercise because of a seizure. Additionally, 36% reported injuries during exercise that were unrelated to seizures and 36% reported

that seizure frequency was reduced as the result of exercise. Conversely, Ablah and colleagues⁴⁵ reported a higher frequency of seizures related to exercise. However, only 53% of the 18% who reported seizures around the time of exercise had a seizure while exercising.

Experimental intervention studies have found that exercise either leads to fewer seizures or does not change seizure frequency. A study examining physical activity in women with intractable epilepsy found that average seizure frequency decreased from 2.9 to 1.7 per week with exercise.⁴⁶ In contrast, Nakken and colleagues demonstrated that a 4-week training program did not alter seizure frequency.⁴⁷ In addition, a longer 12-week study found no significant impact on seizure frequency, but did demonstrate improved vigor, as well as decreased anxiety and depression.⁴⁸ Although these studies do not provide evidence for exercise as a prophylaxis for seizures, the data do not demonstrate that exercise triggers seizures.

The mechanism by which exercise may serve as an antagonist to seizures is not known. Studies that have found a decrease in the frequency of epileptiform discharges on EEGs during exercise^{49–52} suggest that exercise affects the cerebral mechanisms responsible for generating these discharges. During exercise, persons are typically more vigilant, alert, and attentive. Epileptiform discharges can be activated with drowsiness, and studies have documented a decrease in epileptiform discharges when patients with epilepsy are engaged in an interesting task or activity.⁵³ Exercise can also reduce stress and increase endorphin levels.⁵⁴ Stress is a known precipitant for seizures in persons with epilepsy,⁴³ and β -endorphins have been shown to reduce epileptiform discharges.⁵²

Lastly, studies using animal models of epilepsy to investigate the effects of exercise on seizure threshold^{55,56} have suggested other protective effects through mechanisms such as insulin growth factor 1 pathways⁵⁷ or the prevention of oxidative free radical mechanisms.⁵⁸

Although seizure risk with exercise appears to be low, AED compliance is critical because a decrease in serum drug levels increases the risk for seizures in any setting. Variations in drug metabolism could theoretically increase seizure risk (increased metabolism) or increase adverse effects (decreased metabolism). However, changes in drug metabolism have not been observed in studies that have evaluated drug levels during exercise.^{47,48}

ACUTE MANAGEMENT OF SEIZURES

Persons with well-controlled epilepsy may have a seizure during exercise. The first tenet of acute seizure management is to stay calm, because a seizure is almost always a self-limited event that requires minimal intervention. It is important to protect the person from self-injury. Objects close to the person, particularly during a convulsive seizure, should be removed. Restricting equipment or uniforms should be loosened. A tongue blade or other object should never be inserted between the teeth because a person will not swallow his or her tongue. Instead, the person may bite and break the object or injure the care provider. If a mouthpiece is present, it should be removed only if removal can be accomplished safely.

If the seizure does not terminate within 5 minutes, emergency medical services should be activated. Rectal diazepam is often prescribed for persons prone to prolonged seizures. The person administering diazepam should be aware of the potential for decreased respiratory drive. However, the benefit of stopping the seizure generally outweighs the risk of respiratory suppression.⁵⁹

When the seizure is over, the person may be tired and confused. He or she may be allowed to lie down, but because vomiting may occur during the postictal period, he or she should be turned sideways to prevent aspiration. This position is not imperative if the seizure was nonconvulsive (e.g., a focal seizure with alteration of awareness with stereotypic or bizarre behaviors).

After acute management of a seizure, it is important to consider the person's emotional needs. The embarrassment that may be felt should be acknowledged, and it is important to recognize that although some persons may wish to speak about their experience, others may not. A simple "How are you feeling?" is often appreciated. The same approach applies to persons who witness the seizure. Given the many misperceptions about epilepsy,⁶⁰ the witnesses need to know that self-limited seizures are often not harmful, that these persons usually recover, and that most persons lead fulfilling lives with only rare or zero breakthrough seizures.

RETURN-TO-ACTIVITY GUIDELINES

No clear guidelines exist for safe return to activity after a seizure. Activity should be restricted if the person is either lethargic or confused. However, not all seizures result in a prolonged period of postictal symptoms (e.g., absence seizures), and thus return-to-activity decisions should be made on an individual basis.

Although a single self-limited seizure is not harmful, seizures can potentially result in injuries. The most common seizure-related injuries include fractures of the humeral neck, femur, clavicle, and ankle; chipped teeth; and shoulder and hip dislocations.⁴⁴ However, in a study that spanned more than 16 years, Aisenson et al. documented that the accident rate in patients with epilepsy was similar to that of their control patients without epilepsy.⁶¹ In a more recent study by Fischer and Daute, it also was found that during exercise, no difference in accident rates existed in children with or without epilepsy.⁴⁴

Safety Considerations Regarding Participation in Sports

A few factors should be considered to determine eligibility for sports or exercise participation, including (1) seizure types and frequency, (2) seizure precipitants, and (3) the activity in which the person wishes to participate. In general, sports participation should be allowed because restricting participation may have a substantial detrimental impact on a person's health and quality of life. Also, psychological and social responses to epilepsy may induce anxiety, depression, low self-esteem, and peer isolation. Exercise is known to reduce these symptoms.^{62–64}

Participation also should be encouraged because persons with epilepsy may tend to be less active and physically fit. Patients with epilepsy who participate in exercise generally perform the

exercise at less intense levels than control subjects, which is well documented in two studies.^{65,66} In addition, the epilepsy group tended to be overweight.⁶⁶

For most people with good seizure control, few restrictions are recommended. However, notable exceptions exist for persons with frequent or semifrequent atonic (drop) seizures and those who experience frequent or prolonged seizures. For persons recently diagnosed, participation in sports and exercise may need to be delayed until therapy and control of seizures are established.

Sport-specific restrictions are detailed in Box 20.1. Participation in extremely high-risk activities is typically discouraged because even a single seizure during these activities could have dire consequences. For example, a seizure during participation in gymnastics or stunts while cheerleading may significantly predispose a person to injury. Correspondingly, a seizure while climbing monkey bars or trees could have a similar result. In these circumstances, a confident and able “spotter” should be used. In addition, any person prone to seizures should avoid standing near a cliff’s edge or placing themselves in any situation that could cause a fall from a height.

Helmets should always be encouraged for persons who choose to participate in “speed activities” such as bicycling, horseback riding, skateboarding, waterskiing, and snow skiing. With many of these activities, extra supervision is recommended. For persons who choose to participate in motor sports, state laws that guide the “right to drive” need to be considered.⁶⁷ It is important to be aware that sanctioned competitive driving options are also available for children. For this group of children, a frank discussion with the family of the risks versus the benefits, as well as compliance with state laws, is recommended.

Concerns that participation in contact sports, such as football, can exacerbate epilepsy because of repeated head trauma are valid. However, currently no published studies prove that chronic head trauma either increases the frequency of seizures or worsens the epileptic condition.^{60,68–71} In one of the most notable studies previously mentioned, Livingston and Berman⁴² reported their experience with 15,000 young children spanning 36 years. Hundreds of their patients participated in all types of sports, including contact sports such as lacrosse, wrestling, and soccer. Not a single instance of recurrence or aggravation of epilepsy related to head injury occurred in their cohort. Furthermore, it is important to recognize that many athletes with epilepsy have successfully competed at the highest levels of competition, including current NFL players Samari Rolle⁷² and Jason Snelling,⁷³ and professional soccer player Leon Legge.⁷⁴ Nevertheless, as is the case with participation in other sports, the decision to participate in contact sports should be made prudently. The characteristics of the individual’s epilepsy need to be considered because even a single seizure during an inopportune moment could have grave consequences.

Precautions are undeniably necessary during swimming because of an increased risk of drowning.^{75–77} Diekema and colleagues⁷⁷ documented a relative risk of 23.4 for a child (0 to 19 years of age) with epilepsy drowning in a pool. Other studies⁷⁸ have suggested that the risk of drowning for a child with epilepsy is four times that of normal control populations. Nonetheless,

BOX 20.1 Risk of Selected Sports or Recreational Events

High Risk

Bungee jumping
Gymnastics (selected events)
Hang gliding
Motor racing
Mountain or rock climbing
Scuba diving
Skydiving
Stunting
Surfing

Moderate Risk

Bicycling^{a,b}
Horseback riding^{a,b}
Ice skating^{a,b}
Rollerblading^{a,b}
Skateboarding^{a,b}
Sailing/canoeing^{b,c}
Swimming^c
Waterskiing/wakeboarding/wakesurfing^{a,c}

Low or No Risk

Aerobics
Bowling
Cross-country running or skiing^b
Dancing
Field hockey
Golf
Hiking
Track

Contact Sports With No Known Risk

Baseball
Basketball
Football^a
Soccer

^aA helmet should be worn.

^bThe activity should be performed with a partner.

^cSomeone with lifeguard training should be present and aware that the person has epilepsy, a life jacket should be worn, and the activity should not be performed in the ocean or in murky lakes.

these findings do not suggest that persons with epilepsy should not be allowed to swim. With proper precautions, including good seizure control with a stable medication regimen, most persons concur that swimming in a pool or a clear-bottom lake is permitted if the person is well supervised by someone who is trained in rescue and resuscitation and aware of the person’s epilepsy.^{76,79–81} Conversely, swimming in oceans and lakes with murky water is dangerous because in such venues it may be difficult to locate a person who is drowning. Therefore swimming in these environments is discouraged, and if persons choose to swim in murky water, it is essential that a life preserver/vest be used. Additionally, with very few exceptions, scuba diving and competitive diving should be prohibited.^{82,83}

Interestingly, several studies have shown that taking a bath presents a more serious risk for drowning than does swimming.^{76,78,79,84} In a study by Diekema et al.,⁷⁷ children with epilepsy

had a high relative risk of 96 for drowning in a bathtub. In fact, older adolescents and adults run the highest risk because they tend to bathe alone, whereas younger children are usually supervised closely by a parent while bathing.

SUMMARY

Epilepsy does not automatically exclude a person from exercising or participating in sports. However, in making this decision, a balance must always be found between the patient's medical condition and his or her perceived need to participate. Frequently, the health and quality-of-life benefits such as an improved feeling of self-worth, decreased depression, and reduced stress will outweigh the risk of injury resulting from a seizure.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Nakken KO, et al. Effect of physical training on aerobic capacity, seizure occurrence, and serum levels of anti-epileptic drugs in adults with epilepsy. *Epilepsia*. 1990;31:88–94.

Level of Evidence:

IV

Summary:

The study of 21 patients with medically intractable epilepsy demonstrated a largely stable or decreased seizure frequency and stable drug levels during an intensive 4-week training program.

Citation:

Steinhoff B, Neusiiss K, Thegeder H, et al. Leisure time activity and physical fitness in patients with epilepsy. *Epilepsia*. 1996;37(12):1221–1227.

Level of Evidence:

IV

Summary:

This study assessed 136 patients with epilepsy, 145 controls based on a questionnaire, and 35 patients and controls based on standardized clinical tests of physical fitness. The authors concluded that patients with epilepsy show a considerable lack of physical fitness that might have an important impact on their general health and quality of life.

Citation:

Commission of Pediatrics of the ILAE. Restrictions for children with epilepsy. *Epilepsia*. 1997;38(9):1054–1056.

Level of Evidence:

II

Summary:

This is the most recently published guideline by a national medical organization regarding the subject of epilepsy and sports participation. It emphasizes the importance of considering positive effects of activities, such as exercise, before applying restrictions to children with epilepsy.

Citation:

Berg AT, et al. Revised terminology and concepts for organization of seizures and epilepsies: report of the ILAE Commission on Classification and Terminology, 2005–2009. *Epilepsia*. 2010;51(4):676–685.

Level of Evidence:

II

Summary:

These are the most up-to-date recommendations for classifying seizures and epilepsy from the International League Against Epilepsy.

Citation:

Nakken KO. Physical exercise in outpatients with epilepsy. *Epilepsia*. 1999;40(5):643–651.

Level of Evidence:

II

Summary:

This is a survey-based study of 204 adult patients with epilepsy who were compared with the average Norwegian's lifestyle. The authors concluded that for most of the patients, physical exercise had no adverse effects, and for more than one-third, it may have contributed to better seizure control.

Citation:

McAuley JW, et al. A prospective evaluation of the effects of a 12-week outpatient exercise program on clinical and behavioral outcomes in patients with epilepsy. *Epilepsy Behav*. 2001;2:592–600.

Level of Evidence:

II

Summary:

This is a randomized, controlled trial of moderate exercise in 23 patients with epilepsy that demonstrated a positive influence on behavior outcomes with no impact on seizure frequency.

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